Title	**Understanding and Optimizing Content Visibility in Generative Search Engines**
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Project description

Context:

Large Language Models (LLMs) are increasingly changing the way users access information online. Generative search engines and conversational assistants such as ChatGPT, Claude, Gemini or Perplexity can now retrieve information from multiple sources and directly generate an answer, citing only a subset of these sources.

This creates a new research problem for content creators: what makes a source likely to be retrieved, used and cited by a generative engine? The emerging field of Generative Engine Optimization (GEO) studies how online content can be adapted to improve its visibility within such systems.

From a machine-learning perspective, this problem raises several interesting questions. Can we predict whether a document will be selected or cited for a given query? Which characteristics of the query and document are the most predictive? Can such a predictive model subsequently be used to automatically suggest or generate better content?

The project will investigate these questions in a controlled experimental environment. It will build upon existing research on GEO, in particular GEO-bench, a benchmark containing 10,000 queries and associated web sources.

Goals:

The main objective of the project is to build and evaluate models predicting the visibility of a document within a generative search engine.

The first stage will consist in implementing a simplified generative search pipeline inspired by existing literature. Given a user query and a collection of candidate documents, the system will retrieve relevant documents and use an LLM to generate a grounded answer with source citations. This controlled environment will make it possible to generate experimental data and reproduce a subset of existing GEO results and visibility metrics.

The core machine-learning task will then consist in predicting, for a given (query, document) pair, whether the document will be selected and/or cited by the generative engine. Students will first

develop a simple and interpretable baseline based on manually designed features such as semantic similarity, document length, retrieval score or ranking position, combined with standard machine-learning algorithms. More advanced approaches may subsequently be investigated. Examples include transformer-based representations, learning-to-rank methods, interpretability analyses, or LLM-based models.

Finally, the predictive model may serve as the basis for a content optimization system able to identify strengths and weaknesses of a document and suggest modifications expected to improve its visibility while preserving its original meaning.

These extensions are deliberately open-ended: students are not expected to investigate all of them. The exact research direction will be selected according to the first experimental results, progress and interests of the team.

Expected work and deliverables

The project will be organized with increasing levels of difficulty:

- Study the relevant literature and reproduce a simplified version of an existing generative-engine pipeline and GEO evaluation protocol.
- Build an experimental dataset associating queries, candidate documents, retrieval information and citation/visibility outcomes.
- Design and evaluate an interpretable ML baseline predicting document visibility.
- Analyze the model and identify which document/query properties influence visibility.
- Investigate at least one more advanced research direction.

Possible extensions include comparing classical ML with transformer- or LLM-based models; separating retrieval probability from citation probability; automatically optimizing document content; studying robustness to query paraphrases; or measuring whether conclusions obtained with one generative model transfer to another.

If progress permits, experiments may also be conducted with existing commercial generative engines in order to compare the controlled experimental setup with real-world systems. This is considered an optional extension rather than a requirement of the project.

The final deliverables will include the implemented experimental pipeline, reproducible experiments and code, an analysis of the results, and the final report and presentation.

References:

1. Aggarwal, P., Murahari, V., Rajpurohit, T., Kalyan, A., Narasimhan, K. & Deshpande, A. GEO: Generative Engine Optimization. Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, 2024.
2. Liu, N. F., Zhang, T., Liang, P. et al. Evaluating Verifiability in Generative Search Engines. Findings of the Association for Computational Linguistics, 2023.
3. Additional recent literature on generative search, retrieval, learning-to-rank and LLM-based evaluation will be investigated during the project.

Technical aspects

The project will mainly use Python and standard machine-learning/NLP libraries such as scikit-learn, PyTorch and Hugging Face Transformers. Students may use commercial or open-source LLMs through APIs or locally hosted models.

No previous expertise in LLMs or information retrieval is required. The initial baseline is intentionally designed to be accessible using concepts from standard machine-learning courses, while allowing substantially more advanced approaches for interested students.

Comments (e.g. data access, specifics...)

- **Resources made available**

  A subset of existing GEO datasets and benchmarks will be provided or identified at the beginning of the project. Initial code examples and guidance for the controlled generative-search pipeline can also be provided.

  A budget of approximately $500 for LLM API usage will initially be made available and may be increased depending on the research direction and experimental needs.